

Nuvoton NuML_Studio Quick Start with Edge Impulse Project

Description

This document explains how to build a TinyML project with Edge Impulse and NuML_Studio, and deploy it on the NuMaker-M55M1 using image classification as an example.

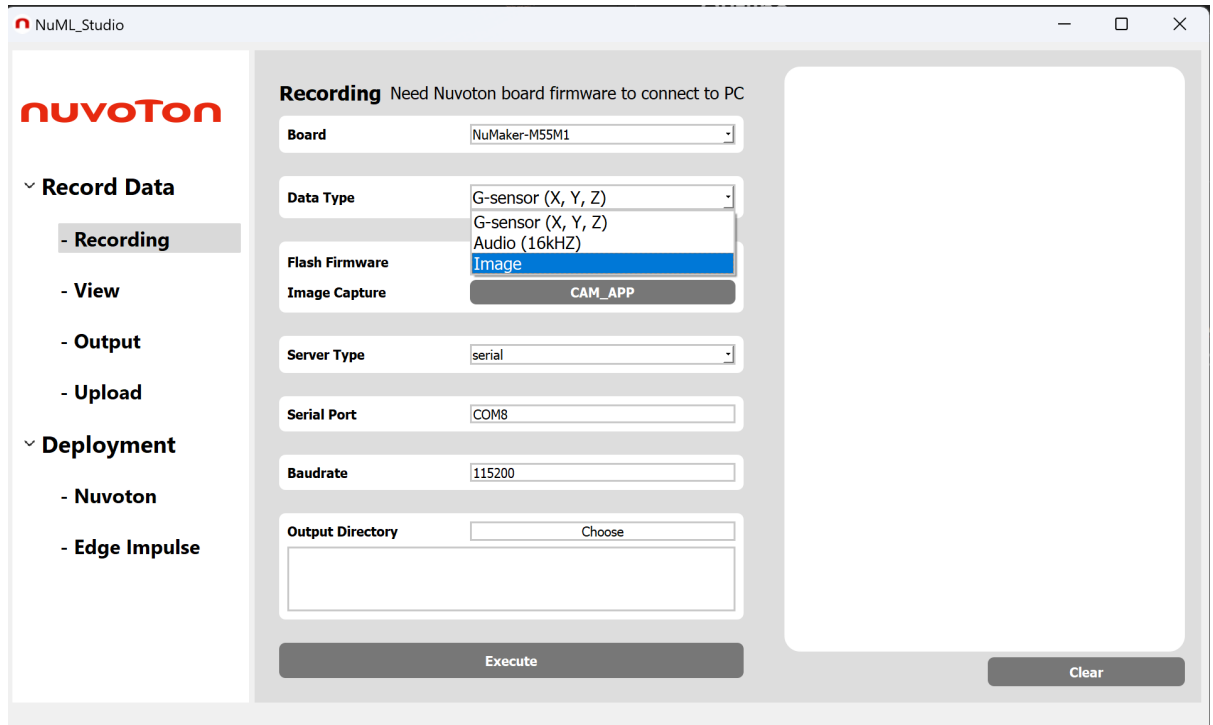
Outline

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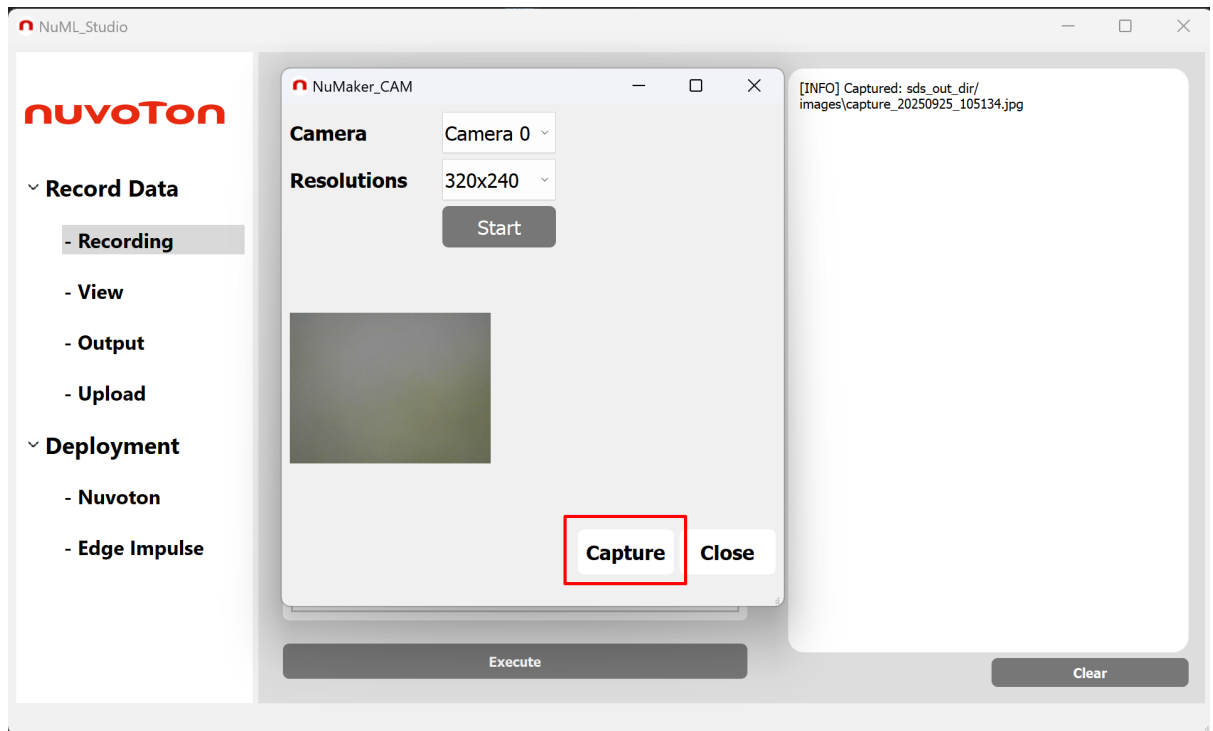
Getting Started

Collecting Data by NuML_Studio

1. Flash the data collection firmware
 - Select **Image**, click **Flash Firmware** and wait for completion.

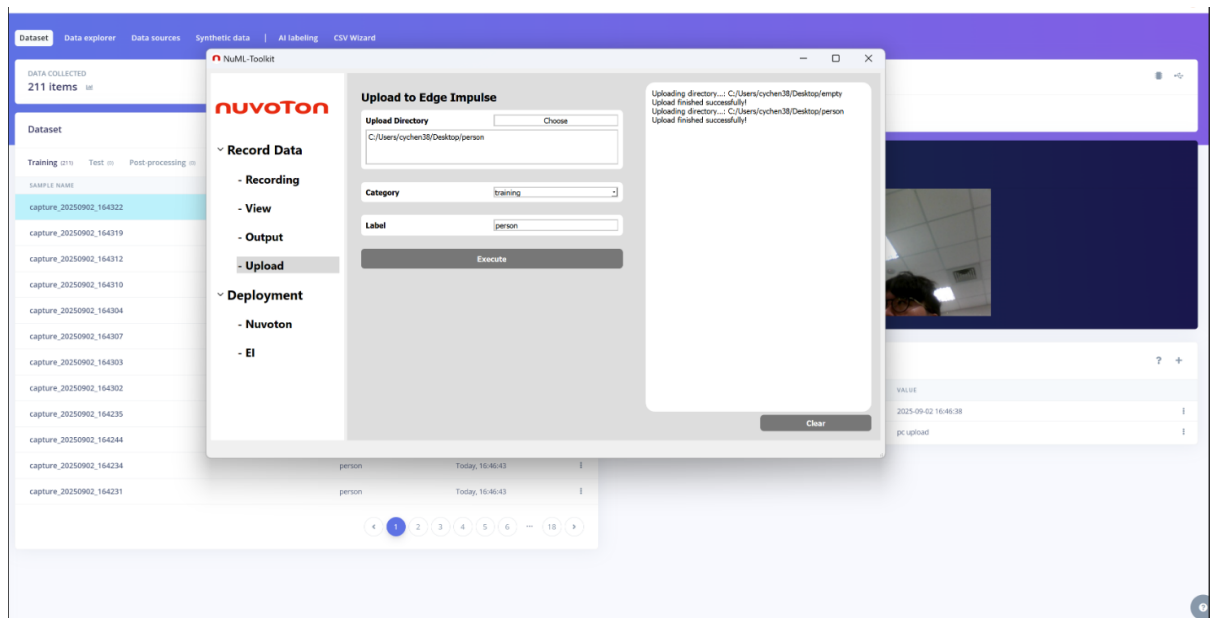


- Click **Image Capture** to take and save images.



2. Upload data to Edge Impulse

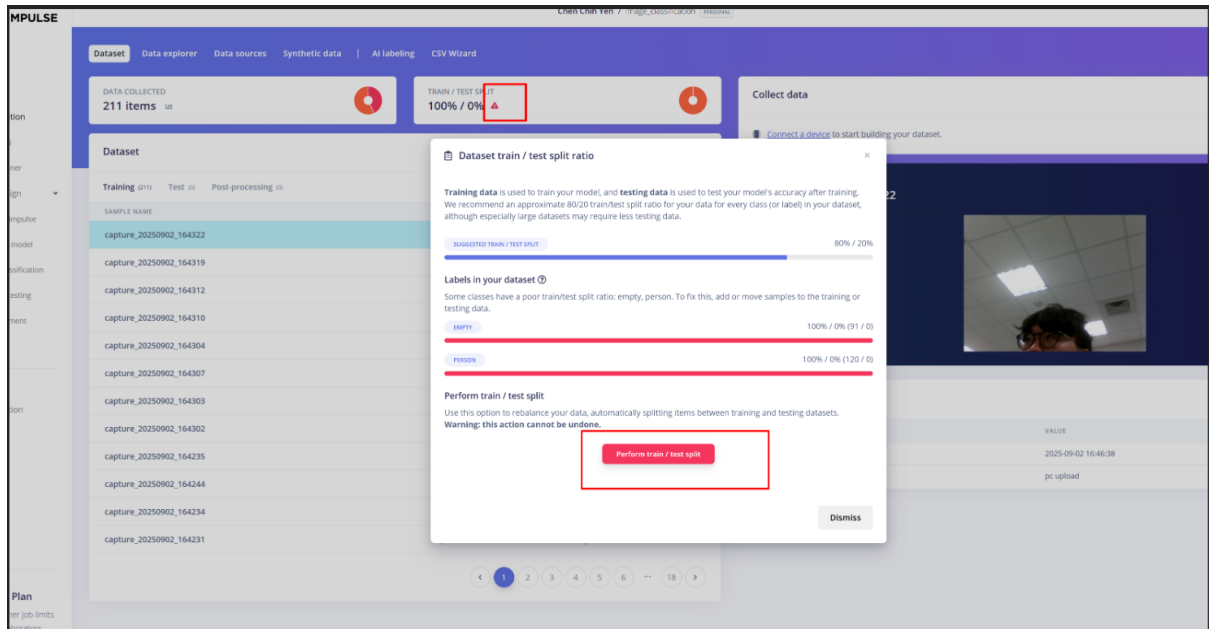
- Copy your project API-Key from Edge Impulse **Dashboard**, **Keys** and paste it into **NuML_Studio_vX.X.X\API_key.txt**.
- Use NuML_Studio to upload your collected data folder, or upload directly via Edge Impulse **Data acquisition** page.



Training by Edge Impulse

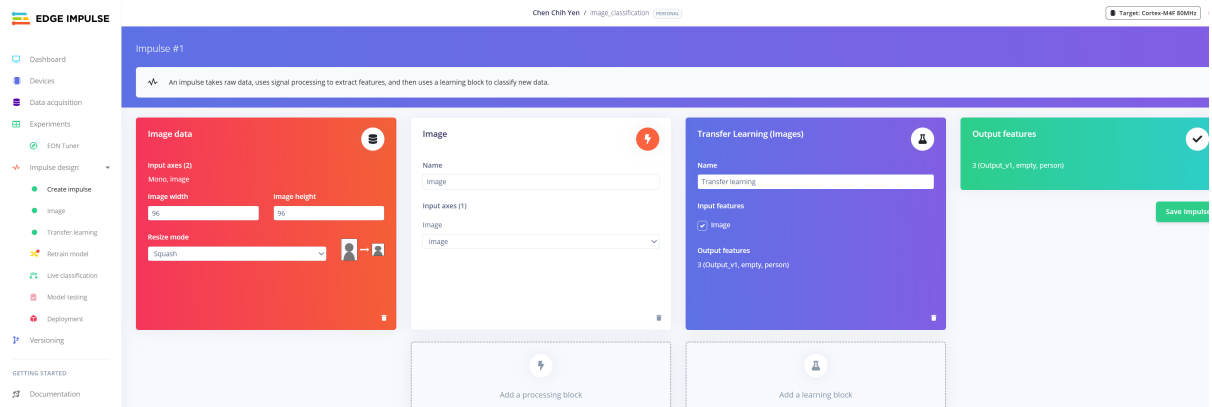
1. Data Acquisition

- Go to the **Data acquisition** tab in Edge Impulse.
- Review and update labels if needed.
- Split the collected data into training and testing datasets.



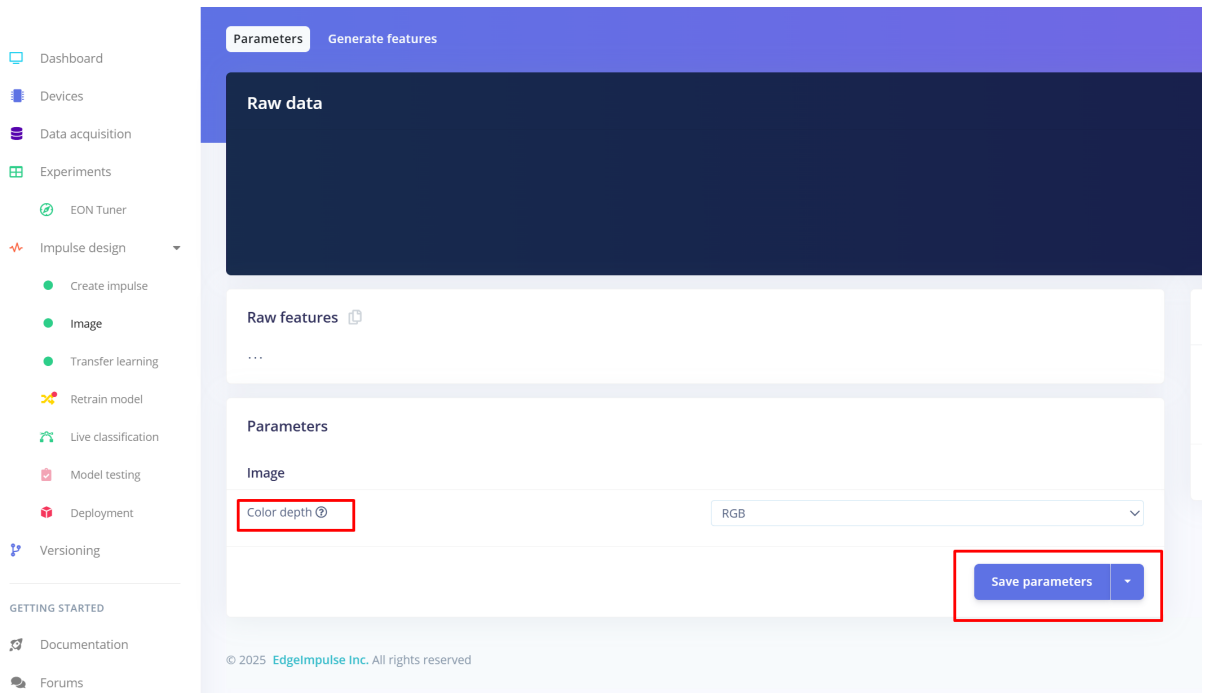
2. Create Impulse

- Go to the **Create Impulse** tab.
- Configure your ML pipeline (preprocessing, learning block, classifier).
- For Image input, set the **Resize mode** to **Squash**

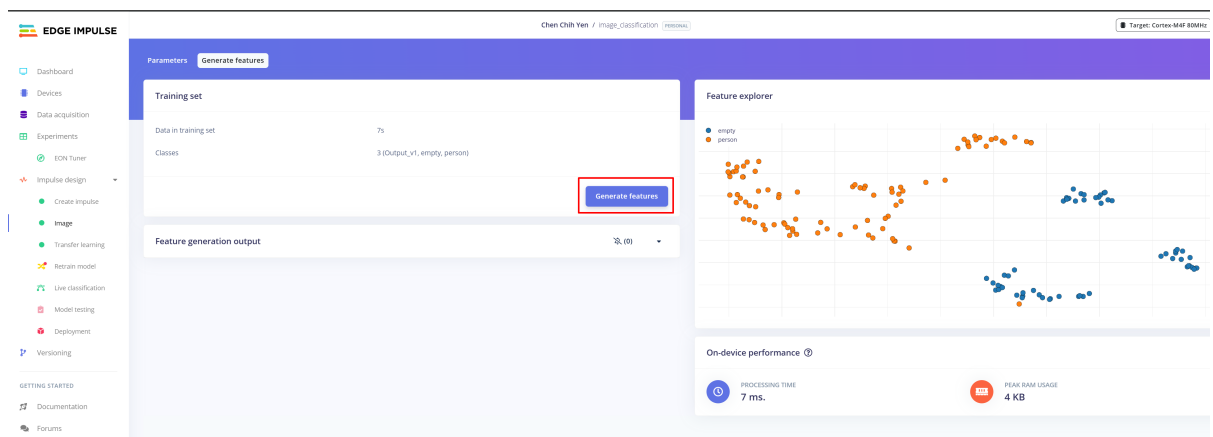


3. Set Parameters and generate Features

- Go to the **Image** tab.
- Set Parameters and click **Save parameters**.



- Then click **Generate features** to extract feature vectors from your dataset.



4. Start Training

- Go to the **Transfer Learning** or **Learning** tab.
- Select the model type.
- Adjust training parameters (epochs, learning rate, validation split, etc.).
- Click **Start training** to begin model training.

Neural Network settings

Training settings

- Number of training cycles: 30
- Use learned optimizer: ☐
- Learning rate: 0.002
- Training processor: GPU
- Data augmentation: ☒

Advanced training settings

- Validation set size: 10 %
- Split train/validation set on metadata key:
- Batch size: 128
- Auto-weight classes: ☒
- Profile int8 model: ☒

Neural network architecture

input layer (27,648 features)

MobileNetV2 96x96 0.1 (final layer: 8 neurons, 0.1 dropout)

Choose a different model

Output layer (2 classes)

Save & train

Training output

Model

Model version: Quantized (int8)

Last training performance (validation set)

ACCURACY: 93.3% LOSS: 0.06

Confusion matrix (validation set)

	EMPTY	PERSON
EMPTY	100%	0%
PERSON	8.5%	91.5%
F1 SCORE	0.95	0.96

Metrics (validation set)

Metric	Value
Area under ROC Curve	0.96
Weighted average Precision	0.95
Weighted average Recall	0.93
Weighted average F1 score	0.94

Data explorer (full training set)

Legend: empty - correct (green), person - correct (blue), person - incorrect (red)

On device performance: EdgeImpulse Compiler (BAAI optimized)

5. Retrain model

- Go to the **Retrain model** tab.

EDGE IMPULSE

Chen Chih Yen / image_classification PERSONAL

Retrain model with known parameters

- Image
- Transfer learning
- Model testing

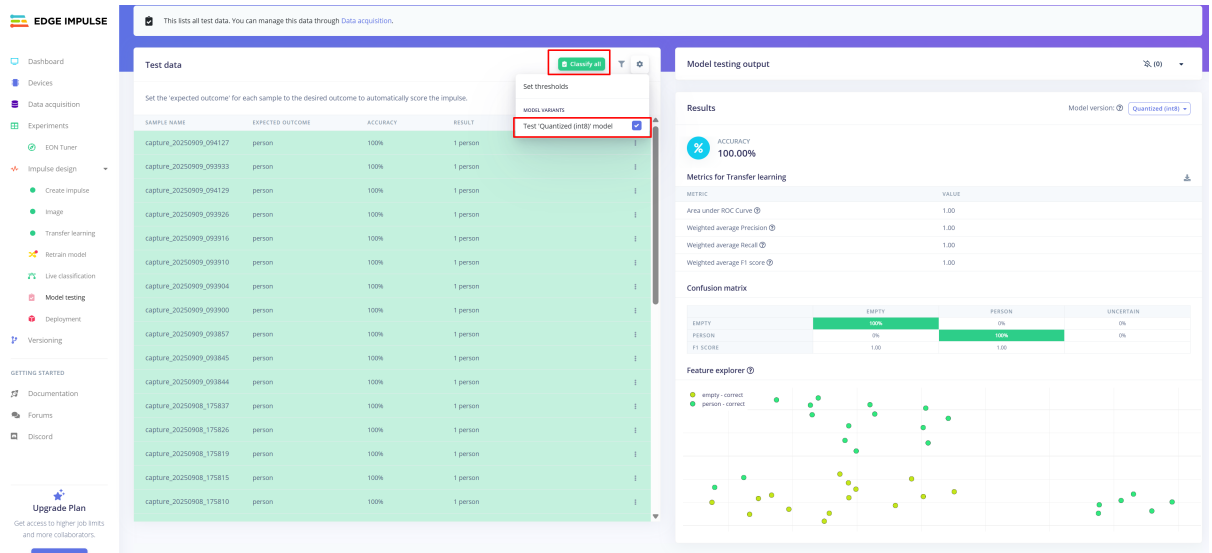
Train model

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Retrain model

6. Test the model

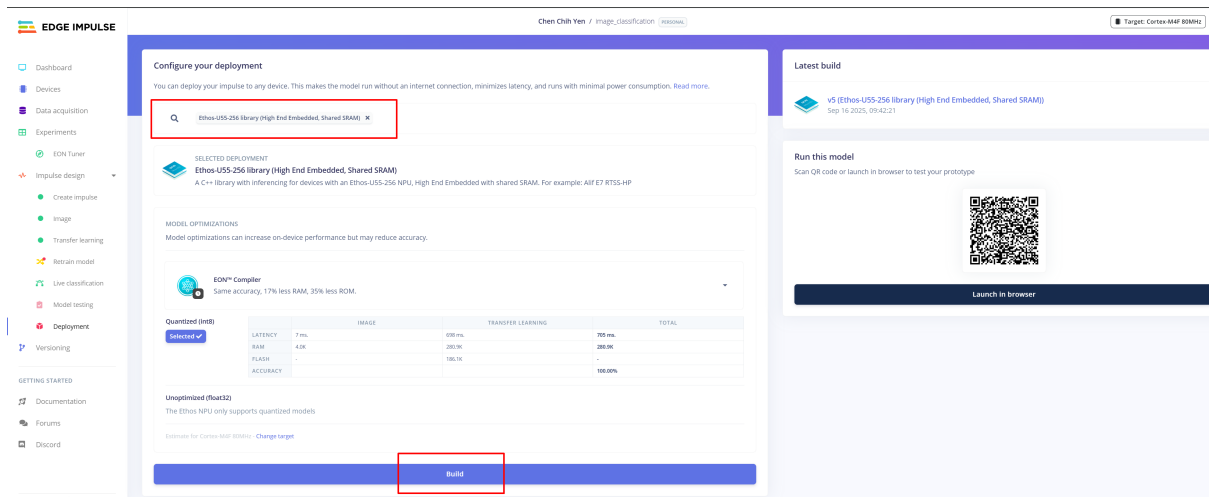
- Go to the **Model testing** tab.
- Please choose **Test Quantized model**. This will test the quantized model using all the test data.



Deployment

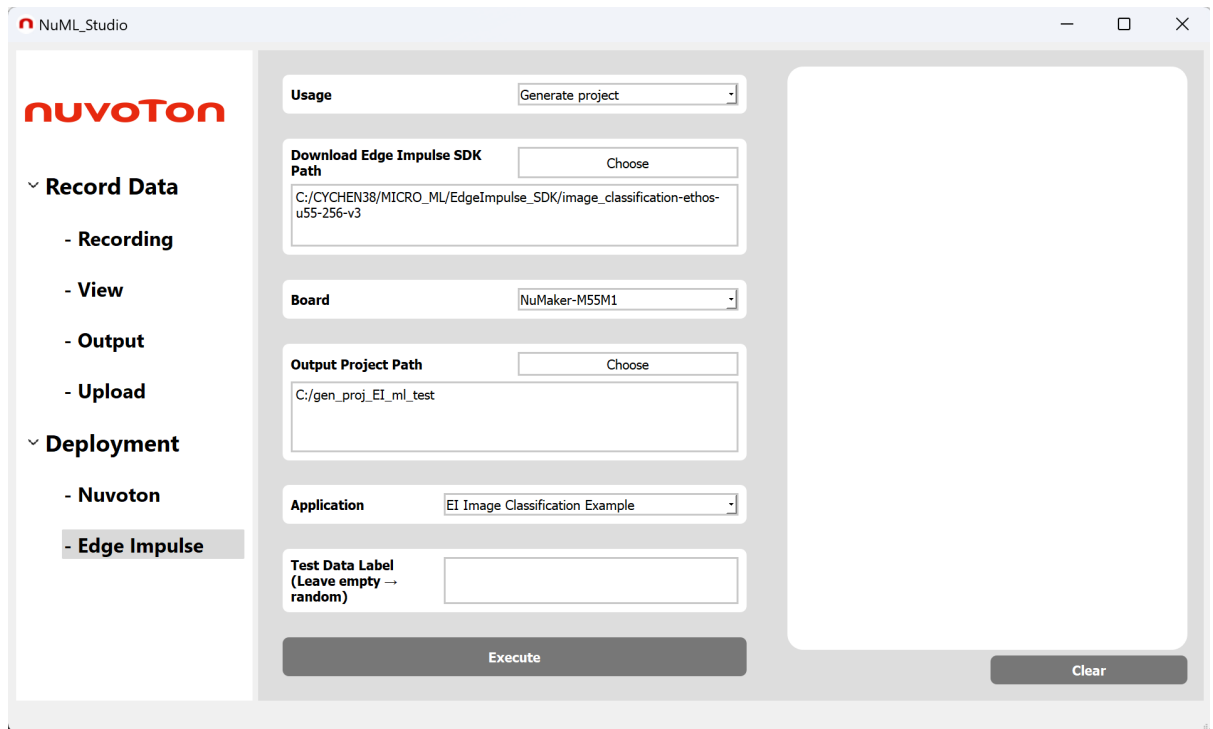
1. Build the deployment model and SDK

- Go to the **Deployment** tab on the Edge Impulse project page.
- Select **ethosu-U55 256 library** and build it.



2. Run inference firmware on NuMaker-M55M1

- After downloading and unzipping the SDK, use NuML-Studio to create the code project.
- In NuML-Studio, select - **Deployment**, - **Edge Impulse**.



- The final step is to open the Keil or VSCode CMSIS project generated by NuML-Studio, then build and flash the NuMaker-M55M1 ML firmware with your Edge Impulse model.